# Lesson 11: Reliability and Validity

*English Stage 4 (Psychology) · Unit 2: Research Ethics and Measurement*

*Adapted from Research Methods in Psychology (4th American edition) by Rajiv S. Jhangiani, I-Chant A. Chiang, Carrie Cuttler and Dana C. Leighton (CC BY-NC-SA 4.0): 4.3 Reliability and Validity of Measurement.*

## In this lesson

You learn how psychologists show that a measure works: its reliability (consistency over time, across items and across observers) and its validity (whether the scores represent what they are supposed to). You practise two features of evaluative academic English: *should* used for expectations, and words that express degree, such as *at best* and *to the extent that*.

## Learning goals

After this lesson you should be able to:

- Explain the three types of reliability and how each is assessed.
- Explain face, content, criterion, convergent and discriminant validity, with the book's examples.
- Decide which kind of evidence a research finding gives about a measure.
- Distinguish *should* of expectation from *should* of advice, and use degree language.
- Plan how to check the reliability and validity of your own measures.

## Key language

| Term or phrase | Meaning | Example |
|---|---|---|
| reliability | The extent to which scores on a measure are consistent across time, across items, and across researchers when there is subjective judgment. | Reliability refers to the consistency of a measure. |
| test-retest reliability | The extent to which scores are consistent across time for the same individuals. | Rosenberg scores a week apart: r = +.95 |
| internal consistency | The extent to which the items on a multiple-item measure are consistent with each other. | People who agree that they are a person of worth should agree that they have good qualities. |
| split-half correlation | The correlation between scores on one half of the items and scores on the other half. | even- vs odd-numbered items: +.88 |
| Cronbach's α | A statistic for internal consistency; conceptually, the mean of all possible split-half correlations. | A value of +.80 or greater indicates good internal consistency. |
| interrater reliability | the extent to which different observers are consistent in their judgments (assessed with α, or Cohen's κ for categorical judgments) | two observers rating social skills from videos |

| Term or phrase | Meaning | Example |
|---|---|---|
| validity | The extent to which scores represent the variable or construct they are intended to; a judgment based on the available evidence. | A measure can be extremely reliable but have no validity. |
| face validity | The extent to which a measure appears "on its face" to measure the construct. | The finger-length method has poor face validity. |
| content validity | The extent to which a measure covers all aspects of the construct. | A test anxiety measure should cover nervous feelings and negative thoughts. |
| criterion validity | the extent to which scores correlate with other variables (criteria) that they should correlate with; concurrent (measured at the same time) or predictive (measured later) | test anxiety and exam performance |
| convergent validity | correlation with existing measures of the same construct | a new and an established test anxiety measure |
| discriminant validity | The extent to which scores are not correlated with measures of conceptually distinct variables. | Self-esteem scores should not correlate highly with mood. |
| to the extent that | as much as; in proportion to how far | To the extent that each participant does have some level of social skills, observers' ratings should be correlated. |
| cast doubt on | make something seem less likely to be true | ... this would cast doubt on the validity of the measure. |

## Rules and explanations

### The ideas

**Demonstrate, do not assume.** Psychologists collect data to show that a measure works, and stop using it if they cannot. If your bathroom scale showed a 10-pound gain after a month of dieting and looser clothes, you would conclude it was broken.

**Reliability** is consistency of three kinds:

| Type | Consistency across | How it is assessed | Book's example |
|---|---|---|---|
| Test-retest | time | the same people measured twice; correlation between the two sets of scores (+.80 or more is good) | Rosenberg scale, a week apart: +.95 |

| Type | Consistency across | How it is assessed | Book's example |
|---|---|---|---|
| Internal consistency | items | split-half correlation, or Cronbach's α (+.80 or more is good) | Rosenberg even vs odd items: +.88; bets in simulated roulette as a measure of risk seeking |
| Interrater | observers | correlation between observers' ratings; α (quantitative) or Cohen's κ (categorical) | social skills rated from videos; counts of aggressive acts with the Bobo doll |

High test-retest correlations are expected only for constructs assumed to be stable (intelligence, self-esteem, the Big Five). Mood changes by nature, so a mood measure with a low test-retest correlation over a month is not a concern. The figures in 4.3 are scatterplots of these two Rosenberg correlations; the text gives the values you need.

**Validity.** Reliability is one kind of evidence, but it is not enough: measuring self-esteem by the length of people's index fingers would be extremely reliable and completely invalid. The "types" of validity are best seen as other kinds of evidence:

- **Face validity**: does it look as if it measures the construct? Usually judged informally and at best very weak evidence, because intuitions are often wrong and many good measures lack it. The MMPI-2 uses 567 statements (the book's "over 567" is a slip), many with no obvious link to what they measure: "I enjoy detective or mystery stories" measures the suppression of aggression. What matters is whether a person's pattern of responses matches that of people who suppress aggression.
- **Content validity**: does it cover the construct? It is checked against the conceptual definition. If test anxiety involves nervous feelings and negative thoughts, the measure needs items on both; a measure of attitudes toward exercise must cover thoughts, feelings and actions.
- **Criterion validity**: do scores correlate with variables they should correlate with? Test anxiety scores should correlate negatively with exam performance and grades, and positively with general anxiety and blood pressure during an exam; physical risk taking with extreme sports, speeding tickets and broken bones. **Concurrent validity**: the criterion is measured at the same time; **predictive validity**: it is measured later.
- **Convergent validity**: do scores correlate with other measures of the same construct? Cacioppo and Petty (1982) showed that scores on their Need for Cognition Scale correlated positively with an academic achievement test and negatively with dogmatism.
- **Discriminant validity**: are scores *not* correlated with conceptually distinct variables? Self-esteem (a fairly stable attitude toward the self) should not correlate highly with mood (how one feels right now). Need for cognition correlated only weakly with cognitive style, and not at all with test anxiety or socially desirable responding.

### Language focus

**1. *Should* of expectation.** In this section *should* often does not give advice; it states what we expect if the measure is valid or reliable:

- "Any good measure of intelligence **should produce** roughly the same scores for this individual next week as it does today." (expectation)
- "Inter-rater reliability would also have been measured in Bandura's Bobo doll study. In this case, the observers' ratings … **should have been** highly positively correlated." (expectation about the past)
- "His measure of test anxiety **should include** items about both nervous feelings and negative thoughts." (advice: what the researcher ought to do)

**2. Expressing degree.** Evaluation needs precise strength words: "**extremely** good test-retest reliability, it would have **absolutely no** validity"; "Face validity is **at best a very weak** kind of evidence"; "they found **only a weak** correlation"; "**To the extent that** each participant does, in fact, have some level of social skills …, different observers' ratings should be highly correlated."

> **Note for Arabic speakers** - Learners often translate *should* with an Arabic word of obligation or advice, and so miss its other meaning. In research writing it often means "is expected to": "Scores should be correlated" is a prediction. - Check prepositions: "correlated **with**", "related **to**" (not "related with"), "consistent **across** time".

## Practice

**1. Reading questions** (section 4.3)

1. What are the three types of reliability, and across what is each a consistency?
2. How is test-retest reliability assessed, and what correlation is generally considered good?
3. Why would a low test-retest correlation for a measure of mood not be a cause for concern?
4. How can a behavioral measure show internal consistency? Give the book's example.
5. What is Cronbach's α, conceptually?
6. Give the book's two reasons why face validity is weak evidence.
7. What is the difference between concurrent and predictive validity?
8. What evidence of discriminant validity did Cacioppo and Petty provide?

**2. Evaluating evidence** (section 4.3 has no exercises; the chapter's exercises are in Lesson 12) A. What kind of reliability or validity evidence does each finding give? Is it good or bad news for the measure?

1. Students complete the Rosenberg scale twice, a week apart; the correlation is +.95.
2. Two observers' counts of each child's aggressive acts with the Bobo doll are highly correlated.
3. A test anxiety questionnaire asks only about nervous feelings, not negative thoughts.
4. Scores on a new physical risk-taking measure predict the number of speeding tickets people receive over the next two years.
5. A new self-esteem measure correlates highly with a measure of current mood.
6. A new test anxiety measure correlates positively with an established test anxiety measure.

B. Invent a measure, like the finger-length example, that would be reliable but not valid. Explain why.

**3. Language** A. In each sentence from 4.3, does *should* express expectation or advice?

1. "People's scores on a new measure of test anxiety should be negatively correlated with their performance on an important school exam."
2. "His measure of test anxiety should include items about both nervous feelings and negative thoughts."
3. "People's scores on a new measure of self-esteem should not be very highly correlated with their moods."

B. Complete with a degree expression from the section: *absolutely*, *at best*, *only a weak*, *to the extent that*.

4. Face validity is _____ a very weak kind of evidence.
5. The finger-length measure would have _____ no validity.
6. They found _____ correlation between need for cognition and cognitive style.
7. A measure of risk seeking is internally consistent _____ participants' bets are consistently high or low across trials.

## Speak with your tutor

- "Let me explain the three types of reliability using the Rosenberg Self-Esteem Scale. Correct my explanation and my prepositions."
- "In PSYC 110 Lecture 13, Professor Bloom calls a 'which superhero are you?' test 'neither reliable nor valid' and an intelligence test based on date of birth reliable but 'absolutely not a valid test'. Compare these with the book's finger-length example."
- "In the same lecture, Professor Bloom gives three kinds of evidence for the Big Five: stability over years, agreement between a roommate, a professor and a mother, and prediction of real-world behavior. Help me name each kind of evidence with the book's terms."
- "Professor Bloom also says that 'something can be valid and not reliable'. The book treats reliability as one kind of evidence for validity. Discuss with me whether a measure with very poor reliability can be valid."

## Writing task

Write a **reliability and validity plan** for the measures you defined in Lesson 10 (sleep and memory in university students, or your own topic): 200-300 words.

- For each measure, say which types of reliability are relevant and how you would assess them. Explain any that are not relevant (for example, is one night's sleep expected to be stable, or does it change, like mood?).
- Give evidence for face and content validity (check the measure against your conceptual definition).
- Name at least one criterion that your scores should correlate with (concurrent or predictive), one measure for convergent validity and one conceptually distinct variable for discriminant validity.

- Use *should* of expectation and at least two degree expressions.

If you work on your own topic, plan the same checks for your own measures.

Self-editing checklist:

- Did I cover all three types of reliability, saying why each does or does not apply?
- Is each criterion something I really expect to correlate with my measure, and in which direction?
- Did I separate convergent (same construct) from discriminant (different construct) evidence?
- Did I use *should* for expectations, not only for advice?
- Are my degree words (*strongly*, *only weakly*, *at best*) precise?
- Did I use "correlated with" and "related to" correctly?

Paste your draft into the chat and ask your tutor for feedback. Then rewrite it.

## Answer key

**1. Reading questions**

1. Test-retest reliability: consistency over time; internal consistency: across items; interrater reliability: across different researchers or observers.
2. Use the measure on a group of people, use it again on the same people later, and compute the correlation between the two sets of scores (usually shown in a scatterplot). A correlation of +.80 or greater indicates good reliability.
3. Mood is not assumed to be stable; by its nature it changes, so a low correlation over a month is expected.
4. If participants make a series of bets in a simulated game of roulette as a measure of risk seeking, the measure is internally consistent to the extent that each person's bets are consistently high or low across trials.
5. The mean of all possible split-half correlations for a set of items (this is how to interpret it, not how it is computed).
6. It is based on people's intuitions about human behavior, which are frequently wrong; and many established measures work well despite lacking face validity.
7. Concurrent: the criterion is measured at the same time as the construct. Predictive: the criterion is measured later, so the scores "predict" a future outcome.
8. Scores correlated only weakly with cognitive style and did not correlate with test anxiety or the tendency to respond in socially desirable ways.

**2. Evaluating evidence** A. 1. Test-retest reliability; good (above +.80). 2. Interrater reliability; good. 3. Content validity; bad, because it does not cover all of the construct as conceptually defined. 4. Criterion validity, specifically predictive validity; good. 5. Discriminant validity; bad: the measure may be measuring mood, not self-esteem. 6. Convergent validity; good. B. (example answer) Measuring intelligence by shoe size: an adult's shoe size gives almost the same score every week, so test-retest reliability is excellent, but shoe size has nothing to do with intelligence, so the measure has no face validity and would not correlate with criteria such as exam grades.

**3. Language** A. 1. expectation 2. advice 3. expectation B. 4. at best 5. absolutely 6. only a weak 7. to the extent that

## My notes